MCC Entrance Test, August 2026 — Day 1

Level T1

Test Information. Level T1 — 90 minutes.
6 problems: 2 Easy (10 pts), 2 Medium (15 pts, two parts), 2 Hard (25 pts, three parts); 100 points total.
Areas: Algebra (×2), Combinatorics (×2), Geometry, Number Theory.
Rules: show your work (partial credit rewards genuine progress); no calculators; read all six before choosing where to start; box your final answers.
A past MCC entrance test, free to try: https://nghia71.github.io/readiness

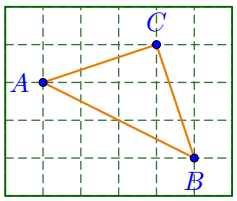
Problem 1 (10 pts). A traveler crosses three toll bridges, one after another. Crossing any bridge doubles the number of coins in the traveler's pouch; immediately afterward, the traveler pays a toll of 24 coins. After paying the toll at the third bridge, the traveler's pouch is completely empty. How many coins did the traveler have before crossing the first bridge?

Problem 2 (10 pts). Two teams, Team A and Team B, each have exactly 20 members and take part in a competition. When the competition ends, every member of Team A shakes hands with every member of Team B, once each. Afterward, every member of Team A also shakes hands with each of their own teammates, once each. If there are n handshakes between the two teams and m handshakes within Team A, what is the value of $n + m$?

Problem 3 (15 pts). A fund starts at exactly 1000 dollars. Each year, on the same date, the fund's value changes by a fixed percentage: Year 1: the value increases by 20%. Year 2: the value decreases by 50%. Year 3: the value increases by 30%. Year 4: the value increases by 40%. What is the fund's value:

(a) right after the Year 2 change takes effect (that is, after the first two changes)? [6 pts]
(b) right after the Year 4 change takes effect (that is, after all four changes)? [9 pts]

Problem 4 (15 pts). On a coordinate grid, a rectangle has corners at $(0, 0)$, $(6, 0)$, $(6, 5)$, and $(0, 5)$. A triangle is drawn inside it with vertices $A = (1, 3)$, $B = (5, 1)$, $C = (4, 4)$.



(a) Find the area of the smallest rectangle, with sides parallel to the grid lines, that completely contains triangle ABC. [4 pts]
(b) Find the area of triangle ABC, and hence find what fraction of the large rectangle's area is covered by the triangle. [11 pts]

Problem 5 (25 pts). A collection of 11 tokens consists of 8 silver tokens, each worth 10 points, and 3 gold tokens, each worth 25 points. A non-empty subset of these 11 tokens is chosen, and its value is the sum of the point-values of the tokens in it.

(a) Suppose the subset uses *only* silver tokens (no gold ones). List every possible value, using at least one silver token. How many different values are possible? [6 pts]
(b) Suppose the subset uses exactly one gold token together with any number of silver tokens (0 to 8 of them). Explain why every value made this way ends in the digit 5, while every value made using only silver tokens ends in the digit 0. [4 pts]
(c) Taking into account ALL non-empty subsets of the 11 tokens (any combination of silver and gold), how many different total values are possible in all? [15 pts]

Problem 6 (25 pts). When you write $\frac{5}{13}$ as a decimal, you get $0.384615384615384615\ldots$, where the block of six digits "384615" repeats forever.

(a) Using long division, verify that $5 \div 13$ produces the repeating block 384615, and explain why the decimal is guaranteed to eventually start repeating. [10 pts]
(b) Using the repeating block, predict what the 15th digit after the decimal point should be — without writing out 15 digits from scratch. Then write the decimal out far enough to check that your prediction is correct. [5 pts]
(c) What is the 2019th digit after the decimal point in the decimal expansion of $\frac{5}{13}$? [10 pts]